

WEEK 9

Information Diffusion

How ideas, behaviors, and viruses spread through networks

C-DE422 Big Data Analytics II

Social Network Analytics

Dr. Amany Eissa | Spring 2026

Where We Are in the Course

Week 9 of 11 — heading into final projects



T O D A Y ' S L E C T U R E

Independent Cascade · Linear Threshold · SIR / SIS · Influence Maximization

Three diffusion mechanics + the optimization problem they motivate.

Learning Objectives

By the end of this lecture you will be able to...

- 1 Distinguish between mechanics and optimization in diffusion problems
- 2 Trace cascades by hand under both Independent Cascade and Linear Threshold models
- 3 Explain why IC is stochastic and LT is deterministic, and what that means
- 4 Identify when SIR vs SIS is the appropriate epidemic framework for a given problem
- 5 Articulate why high-degree seeding is suboptimal, and how greedy seed selection improves on it

PART 1

Why Diffusion Matters

A Thread Running Through the Whole Course

We have been quietly studying diffusion since week 1

Week 1 – 2

Networks as channels

Networks as systems through which news, rumors, and behaviors travel; bridges carry the spread.

Week 3

Clustering

High clustering = efficient local sharing. Low clustering = spread relies on hubs.

Week 4

Betweenness

Important nodes are bridges over which information flows between groups.

Week 5 – 6

PageRank

Influence flows recursively through followers; PPR measures relevance from a seed.

Week 7

Communities

Within-community ties dense; between-community ties are diffusion channels.

Week 8

Link Prediction

Predicting future edges — the channels that future cascades will travel along.

Today: we make the spreading process itself the object of study.

Why We Care: Four Real-World Stakes

Every domain below is a diffusion problem in disguise



Viral Marketing

Pick a small set of seed customers who will spread a product. Wrong seeds → no traction.



Epidemic Control

Predict where infection spreads, decide who to vaccinate first, estimate when peak hits.



Misinformation

Track how rumors spread, identify super-spreaders, design interventions that slow them.



Behavior Adoption

Why do some products go viral while better ones fail?
Why does one neighborhood adopt solar?

Mini Case Study: Hotmail (1996)

The first famous viral-marketing diffusion success

THE PROBLEM

Hotmail launched in 1996 with a tiny marketing budget and needed to compete against established email providers. They could not afford TV or print ads.

THE TRICK

Every outgoing email automatically appended: "PS: I love you. Get your free email at Hotmail."

THE RESULT

12 million users in 18 months

Total marketing spend: ~\$50,000

Acquired by Microsoft for \$400M in 1997.

WHY THIS IS A DIFFUSION PROBLEM

Each new user becomes a sender → spreads to their contacts
→ some sign up → repeat.

This is exactly the cascade dynamic we will formalize today.
The key questions:

- **How fast does it spread?**
- **How big does the cascade get?**
- **Could we have predicted takeoff?**
- **Who should we have seeded?**

PART 2

Two Big Questions

The Field Splits Into Two Questions

Today we touch both, but most of the lecture is about mechanics

MECHANICS

How does spreading work?

Given a network and a starting set of active nodes, predict the size and shape of the cascade.

Models we will study:

- Independent Cascade (IC)
- Linear Threshold (LT)
- SIR / SIS epidemic models

OPTIMIZATION

Who should we seed?

Given a budget of k seeds, find the seed set that maximizes the expected cascade.

Topics:

- The Influence Maximization problem
- Why simple heuristics fail
- The greedy algorithm

You need mechanics before optimization — you cannot maximize what you cannot predict.

Common Setup for All Diffusion Models

Vocabulary you will need for the rest of the lecture

■ Network $G = (V, E)$

- Set of nodes V and edges E . May be directed or undirected.

■ Node states

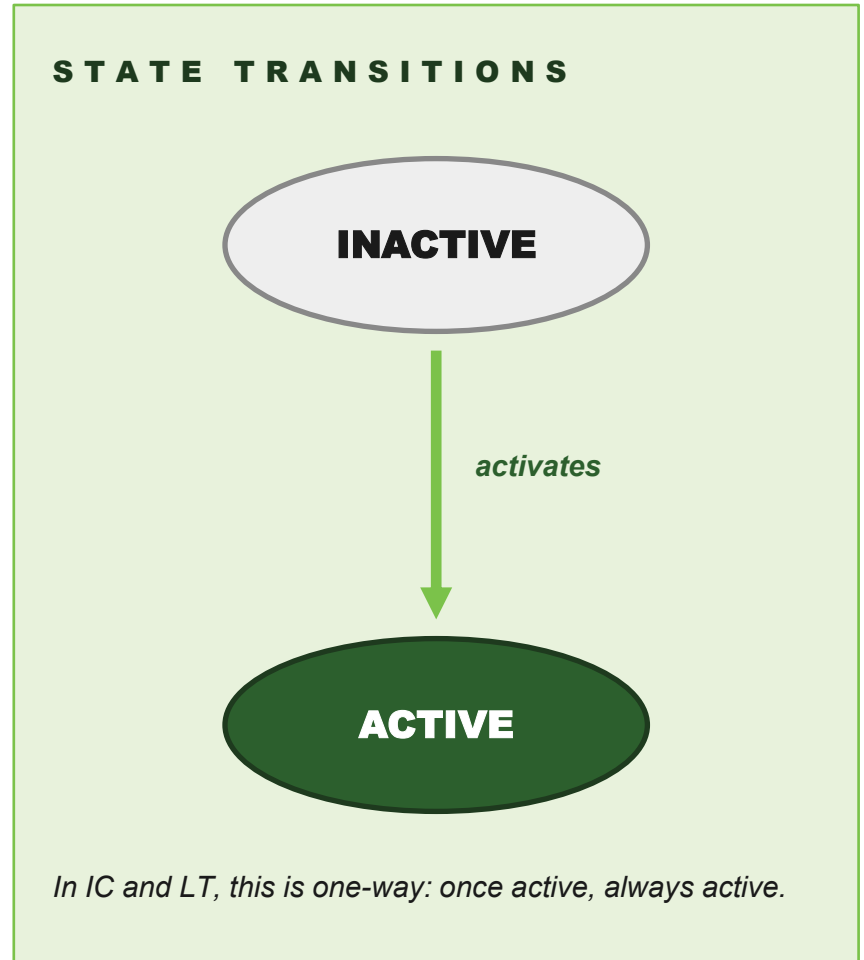
- ACTIVE (or infected, adopted, informed)
- INACTIVE (susceptible, has not adopted)
- Some models add: RECOVERED, EXPOSED, etc.

■ Seed set S_0

- The nodes that start active at time $t = 0$.

■ Cascade size $\sigma(S_0)$

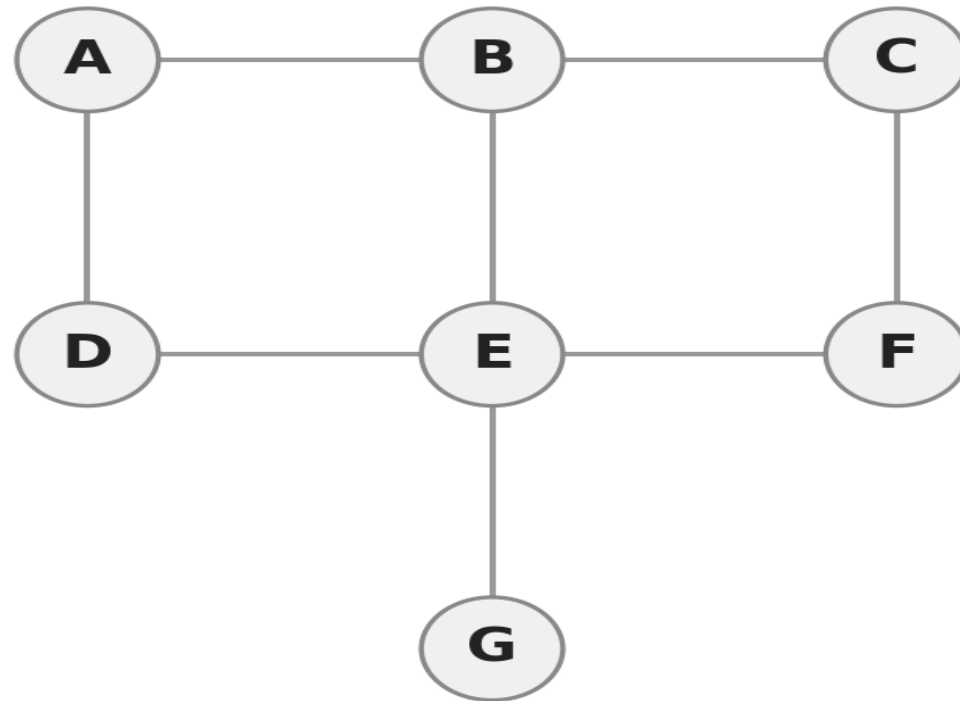
- The total number of active nodes when the process ends.



Our Worked-Example Graph

Same 7-node network used for IC, LT, and influence maximization

The Network — 7 nodes, 8 edges



Degrees: $A=2$, $B=3$, $C=2$, $D=2$, $E=4$, $F=2$, $G=1$

KEY FACTS

7 nodes, 8 edges

Single connected component

Hub: node E (degree 4)

Periphery: G (degree 1)

Why this graph?

Small enough to trace by hand. Has a hub, a periphery, and a triangle-free structure that makes thresholds matter.

PART 3

Independent Cascade Model

The Independent Cascade (IC) Model

Goldenberg, Libai & Muller, 2001

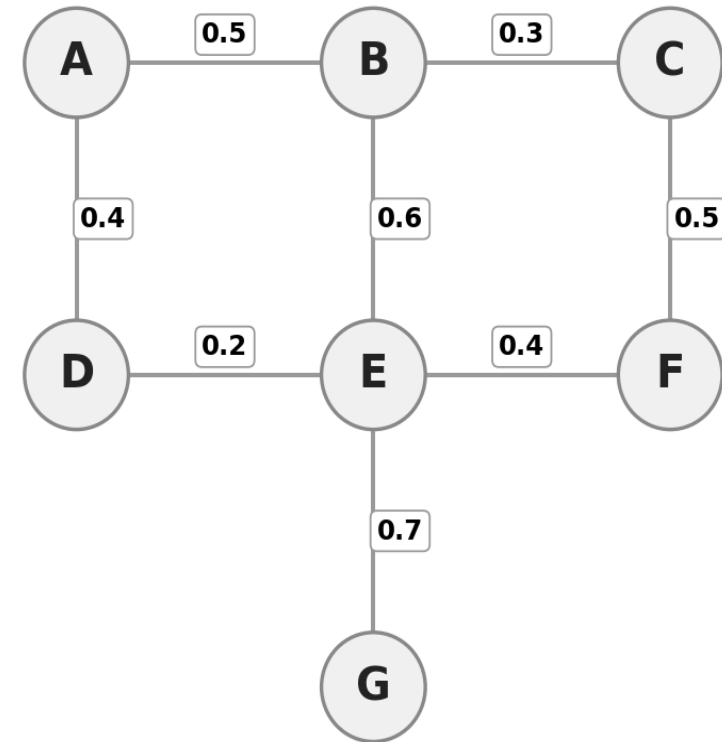
THE IDEA

Each newly active node u gets one chance to activate each inactive neighbor v . Whether u succeeds is decided by an independent coin flip with success probability $p(u,v)$.

THE MECHANICS

- At $t=0$, the seed set S_0 is active.
- When u becomes active at time t , in step $t+1$ it tries to activate each currently inactive neighbor v exactly once.
- Success: u activates v with probability $p(u,v)$. Failure: v stays inactive — and u never tries v again.
- Process ends when no new activations happen in a round.

Network with Edge Activation Probabilities (IC Model)



Two Crucial IC Rules That Trip Up Beginners

Internalize these or your traces will be wrong

1

ONE-SHOT RULE

A failed activation attempt is permanent. If u tries v and the coin says "fail," u never gets to try v again — even if u stays active forever.

WHY IT MATTERS

Order of attempts matters. By the time an inactive node v finally activates, several of its neighbors may have already tried and failed — those channels are closed forever.

2

INDEPENDENCE RULE

Every coin flip is independent of every other. Multiple newly-active neighbors of v each get a separate, independent attempt to activate v in the same round.

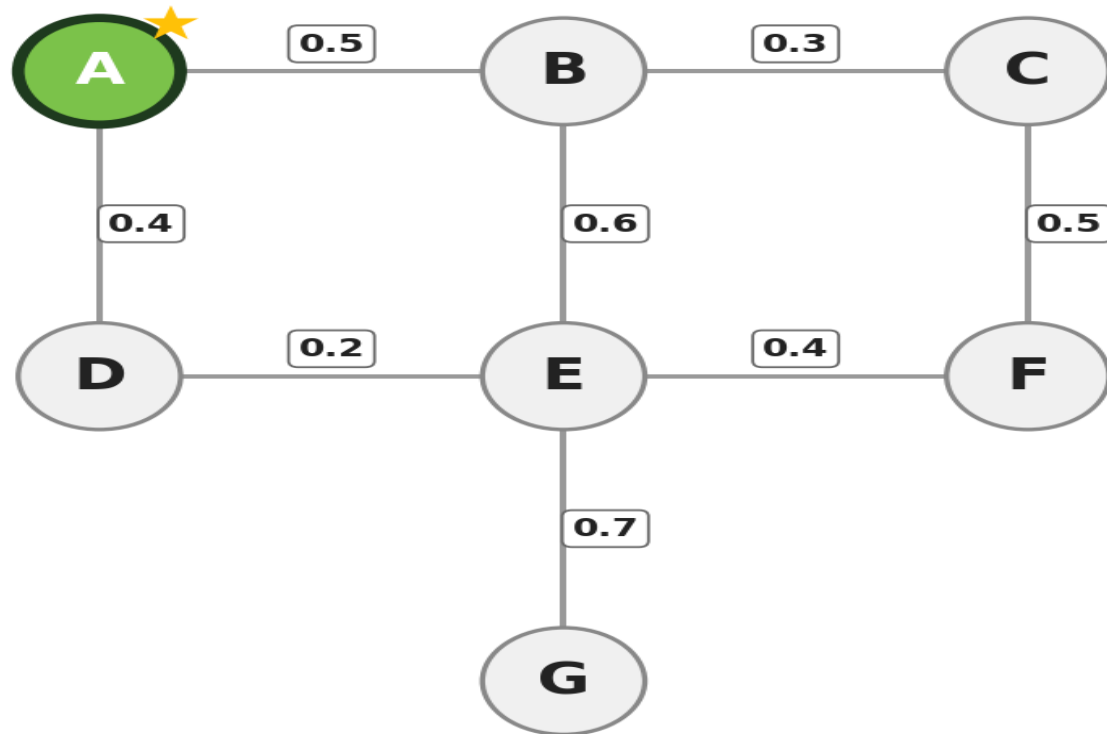
WHY IT MATTERS

If two neighbors u_1 and u_2 both try to activate v , v becomes active if at least one succeeds. So pressure builds: more active neighbors → higher chance v turns.

Worked Example — IC on Our Graph

Seed = {A}. Edge probabilities shown. We use pre-specified coin flips for a deterministic classroom trace.

IC — Round 0 (Initial Seed)



LEGEND

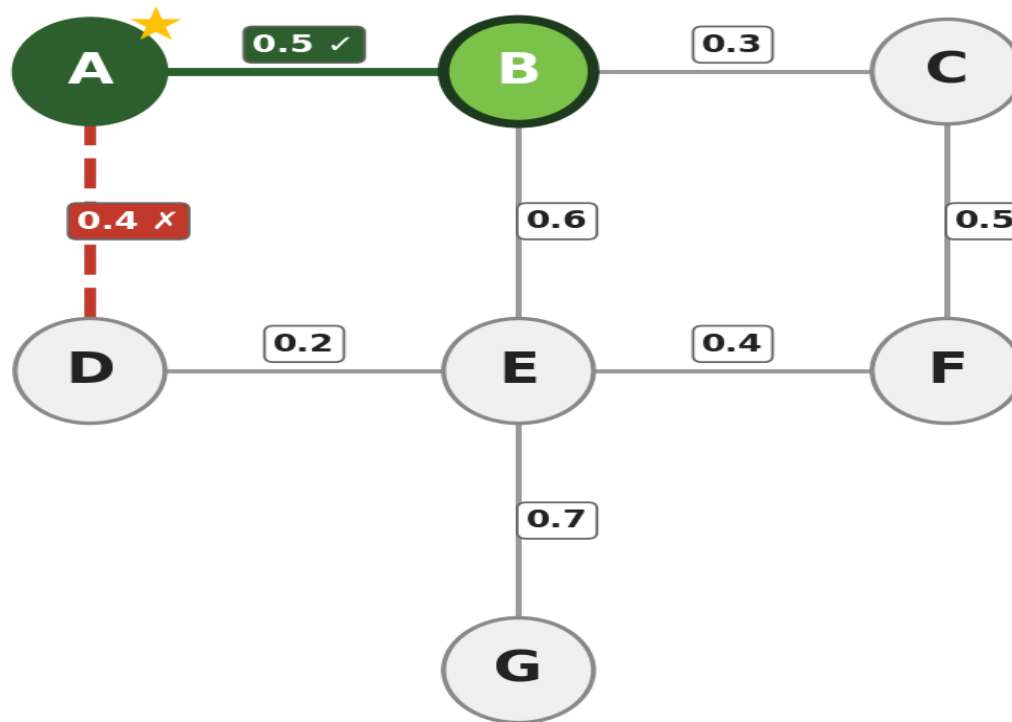
- Active
- Newly active
- Inactive
- Seed
- success
- failed try

How is success/fail decided in practice? Draw a uniform random number $r \in [0, 1]$. Success if $r \leq p$, fail otherwise. We use **fixed coin outcomes** here so the trace is reproducible — real runs vary.

IC — Round 1: A attempts its neighbors

A is active. A's inactive neighbors are B and D. A flips a coin for each.

IC — Round 1: A attempts neighbors



STEP-BY-STEP

A → B

$p = 0.5$, coin = SUCCESS

B activates this round.

A → D

$p = 0.4$, coin = FAIL

D stays inactive. By the one-shot rule, A will NEVER try D again.

End of round 1:

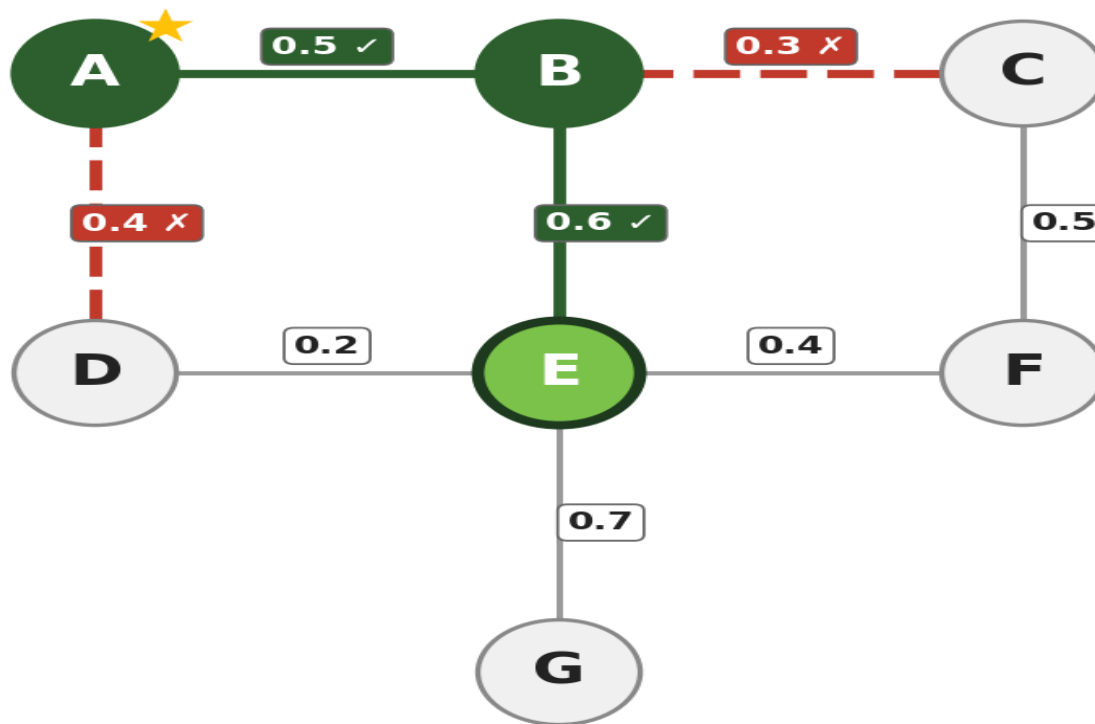
Active = {A, B}

A→B: ✓ activates B | A→D: ✗ fails (D not active, never retried by A)

IC — Round 2: B attempts its neighbors

B (newly active) tries each inactive neighbor: C and E

IC — Round 2: B attempts neighbors



B→C: ✗ fails (C lost forever from B) | B→E: ✓ activates E

STEP-BY-STEP

B → C

$p = 0.3$, coin = FAIL

C lost to B forever.

B → E

$p = 0.6$, coin = SUCCESS

E activates this round.

Note: A does NOT try anyone this round. Only newly-active nodes attempt.

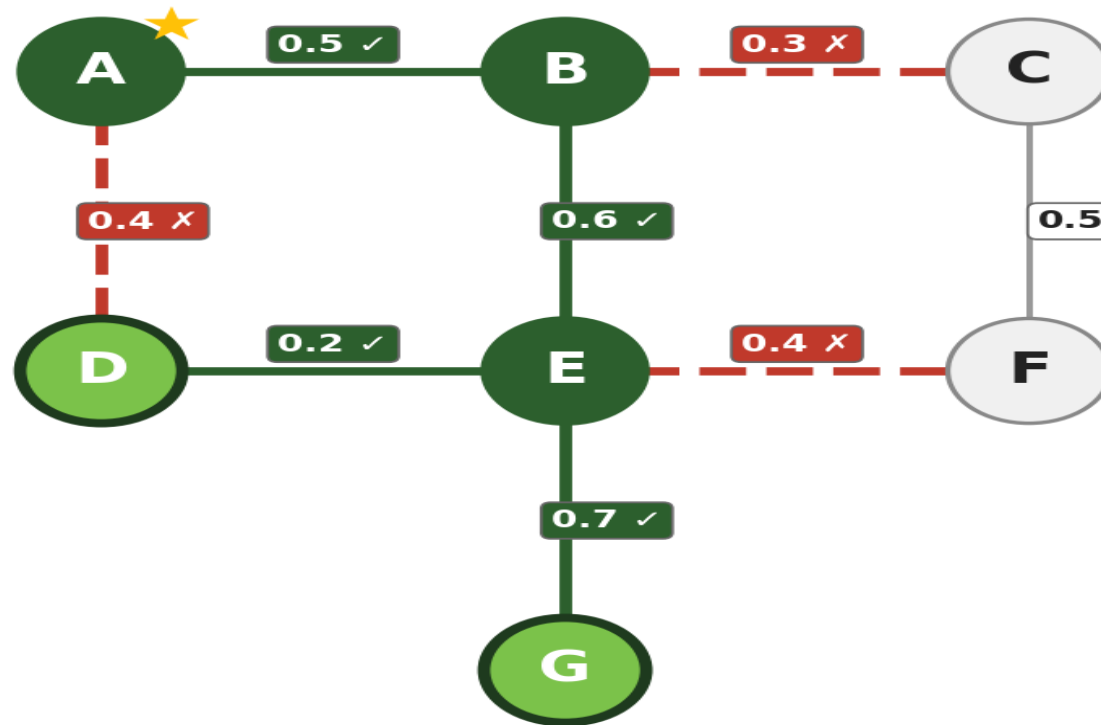
End of round 2:

Active = {A, B, E}

IC — Round 3: E attempts its neighbors

E (newly active) tries D, F, and G

IC — Round 3: E attempts neighbors



E → D ✓ | E → F ✗ (E's only chance at F) | E → G ✓

STEP-BY-STEP

E → D

$p = 0.2$, SUCCESS ✓

D activates via E (a different path than $A \rightarrow D$, which failed).

E → F

$p = 0.4$, FAIL ✗

E → G

$p = 0.7$, SUCCESS ✓

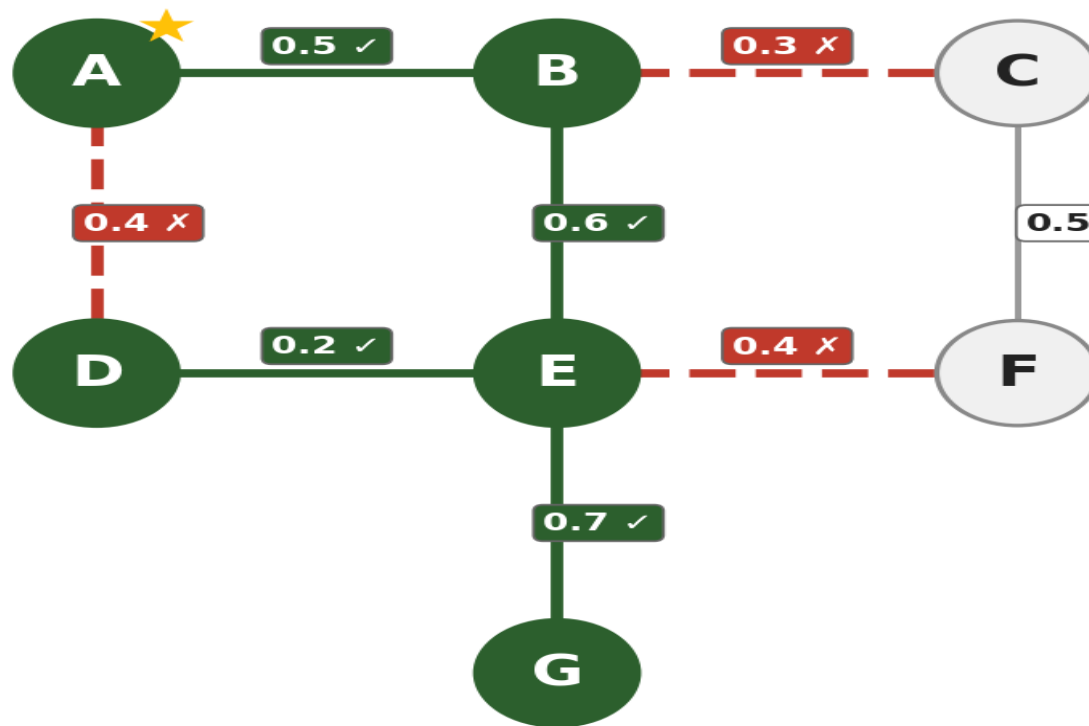
End of round 3:

Active = {A, B, D, E, G}

IC — Cascade Ends

D and G have no inactive neighbors → no more attempts → cascade ends

IC — Final State (cascade ends)



FINAL OUTCOME

5 / 7

nodes activated

ACTIVE

A, B, D, E, G

INACTIVE

C, F

← *lost to failed attempts*

Active = {A, B, D, E, G} Inactive = {C, F} Cascade size = 5/7

IC Is Stochastic — Different Runs Give Different Outcomes

We just saw ONE realization. The same seed could produce many different cascades.

WHAT THIS MEANS

With the same graph, same probabilities, same seed {A}, we could get cascades of size 1, 2, 3, ... up to 7. Each outcome has some probability.

Lucky run

7/7

All coins land right → full network activated

Our run

5/7

Mixed outcomes — what we just traced

Unlucky run

1/7

A→B and A→D both fail → cascade dies immediately

Key consequence: *to compare seed sets, we run many simulations and compare the EXPECTED cascade size.*

PART 4

Linear Threshold Model

The Linear Threshold (LT) Model

Granovetter, 1978 — different intuition from IC

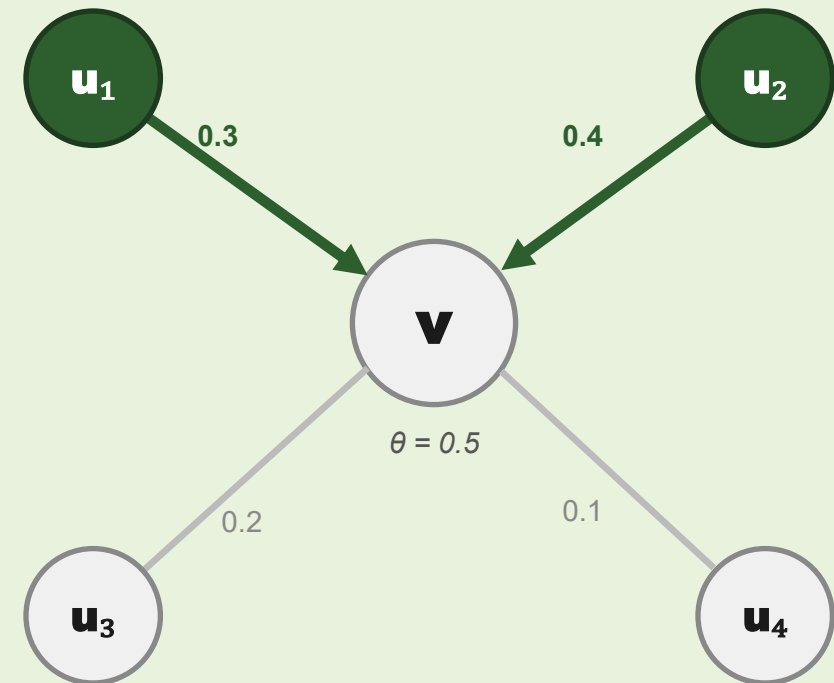
THE IDEA

A node v adopts when ENOUGH of its neighbors have already adopted. Each neighbor exerts a fixed amount of influence; v has a personal threshold for how much pressure it needs to flip.

THE INGREDIENTS

- **Edge weight $b(u,v)$**
 - How much influence neighbor u exerts on v . Sum over u must be ≤ 1 .
- **Threshold $\theta_v \in [0, 1]$**
 - How much total pressure v needs in order to activate.
- **Activation rule**
 - v activates when $\sum b(u,v) \geq \theta_v$, summed over active neighbors u .

INTUITION: PEER PRESSURE



Pressure on $v = 0.3 + 0.4 = 0.7 \geq \theta = 0.5 \checkmark$

v activates

LT vs IC — How Are They Different?

Same diffusion theme, very different mechanics

IC

Sender-driven | Stochastic

- Active node u rolls a coin to push to each neighbor
- Each push is one-shot — fail and that channel is closed
- Random outcome — same setup gives different runs
- Mental model: pandemic, viral content, gossip
- Multiple active neighbors $\rightarrow$ multiple independent shots at v
- Best for: contagion-like, low-commitment spreading

LT

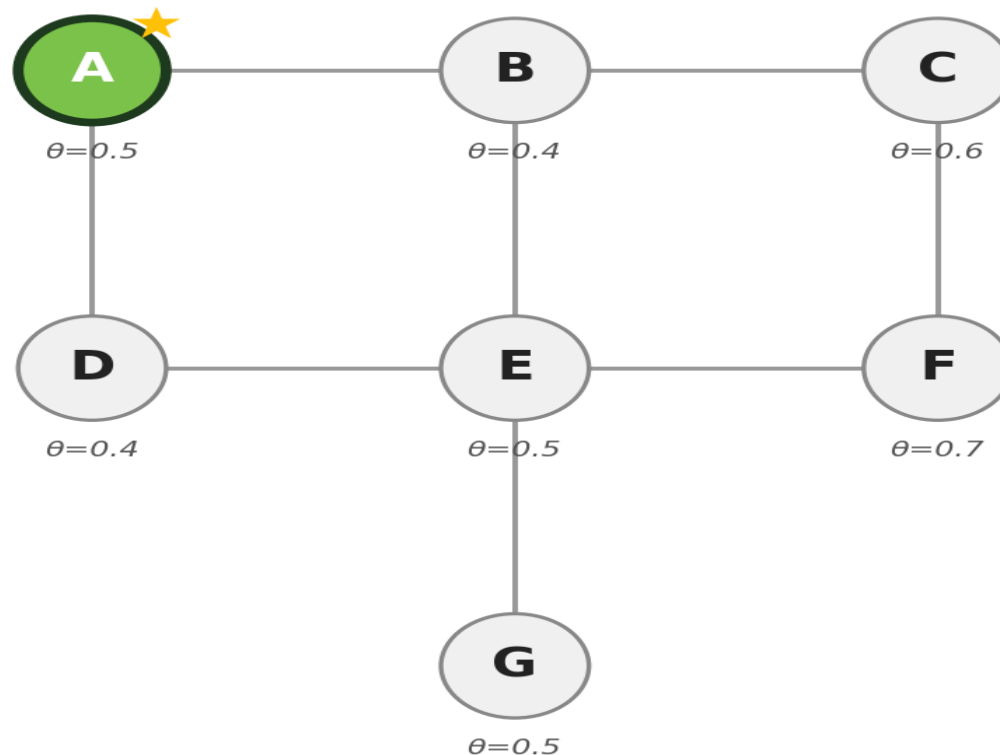
Receiver-driven | Deterministic

- Inactive node v sums up pressure from active neighbors
- v flips the moment cumulative pressure $\geq \theta_v$
- Deterministic given thresholds — same setup gives same result
- Mental model: technology adoption, peer pressure, social proof
- Pressure literally adds: $\text{pressure}(v) = \sum b(u,v)$ over active u
- Best for: decisions requiring conviction or social validation

LT Setup on Our Graph

Standard textbook setup: weight from u to v is $1/\deg(v)$ — incoming weights at v sum to exactly 1

LT — Round 0: Seed = {A}, all thresholds shown



THRESHOLDS

A $\theta = 0.5$ (deg 2)

B $\theta = 0.4$ (deg 3)

C $\theta = 0.6$ (deg 2)

D $\theta = 0.4$ (deg 2)

E $\theta = 0.5$ (deg 4)

F $\theta = 0.7$ (deg 2)

G $\theta = 0.5$ (deg 1)

Each neighbor of v contributes $1/\deg(v)$ of pressure.

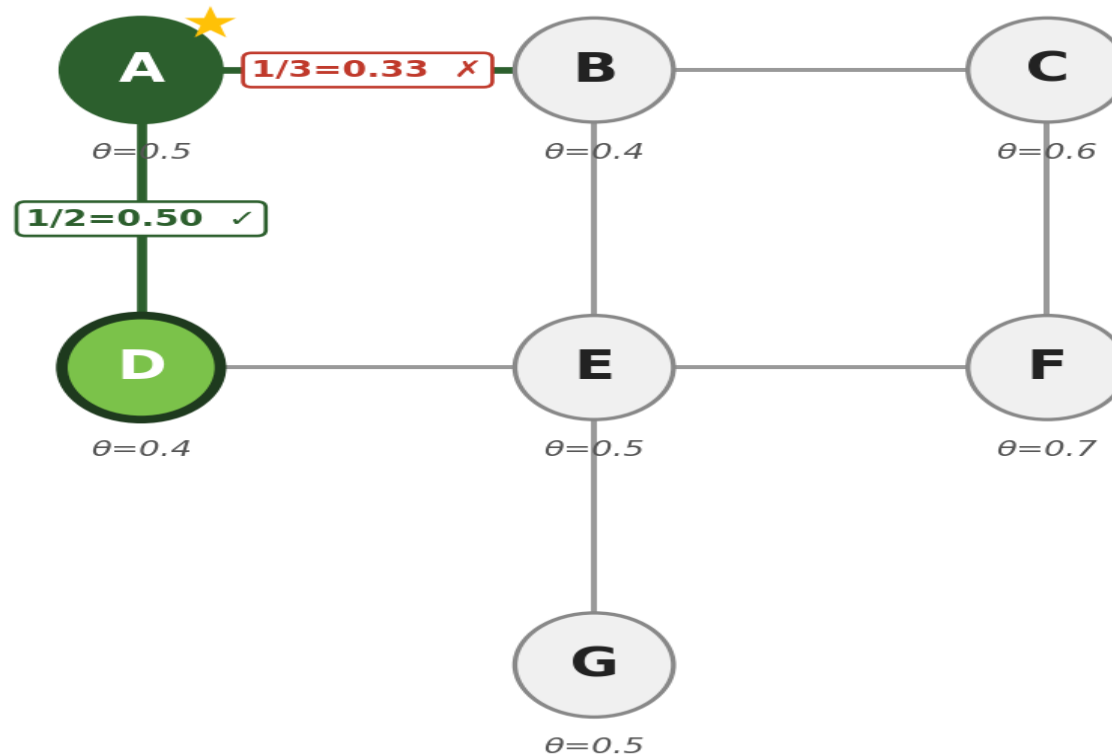
Seed = {A}. Same seed as IC.

Where do thresholds come from? Drawn randomly in theory studies, learned from data in empirical work, or — as here — chosen for a clean classroom example.

LT — Round 1: Compute pressure on each inactive node

Active = {A}. For each inactive v , count active neighbors and divide by $\deg(v)$.

LT — Round 1: Compute pressure on each inactive node



B: $1/3 = 0.33 < \theta=0.4$ x | **D: $1/2 = 0.50 \geq \theta=0.4$ ✓ ACTIVATES**

PRESSURE CHECK

B: 1 active nbr / deg 3
= 0.333 vs $\theta = 0.4$
x Not enough pressure

D: 1 / 2 = 0.500
vs $\theta = 0.4$
✓ D ACTIVATES

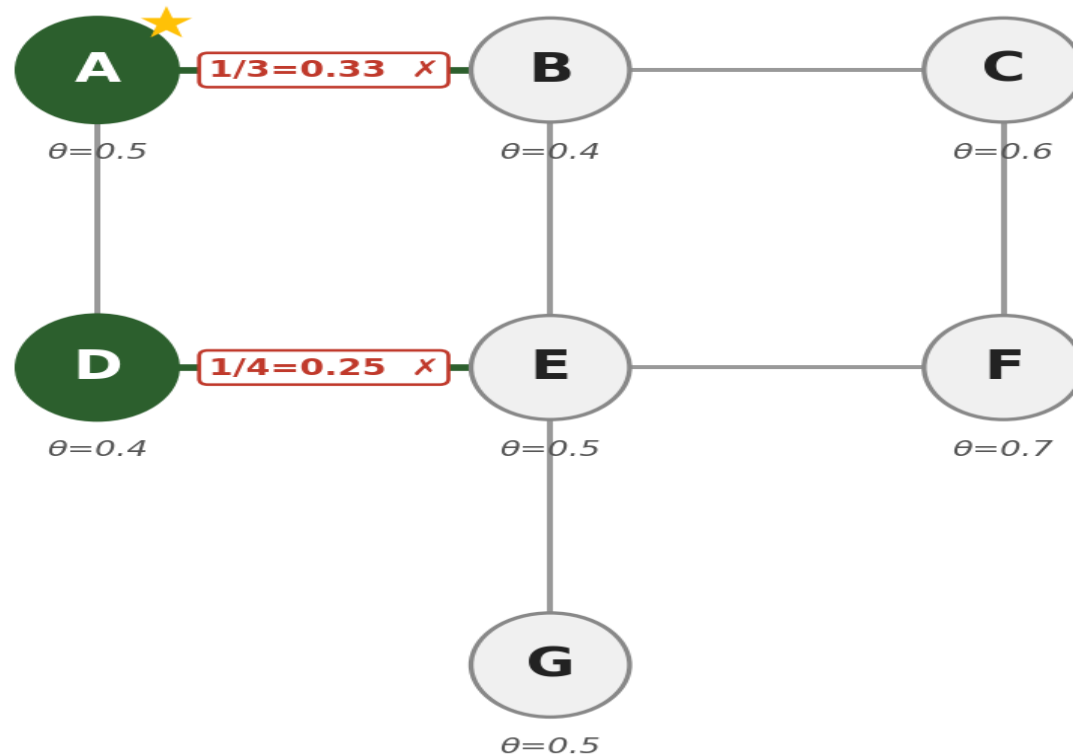
C, E, F, G:
no active neighbors yet

End of round 1:
Active = {A, D}

LT — Round 2: Recheck. Cascade ends.

D is now active. Recompute pressure on every still-inactive node.

LT — Round 2: Re-check (D is now active)



RECHECK

B: still $1/3 = 0.333$
(D is not B's neighbor)
X still below 0.4

E: $1/4 = 0.25$ (from D)
vs $\theta = 0.5$
X still below threshold

Nobody flips this round

→ **CASCADE ENDS**

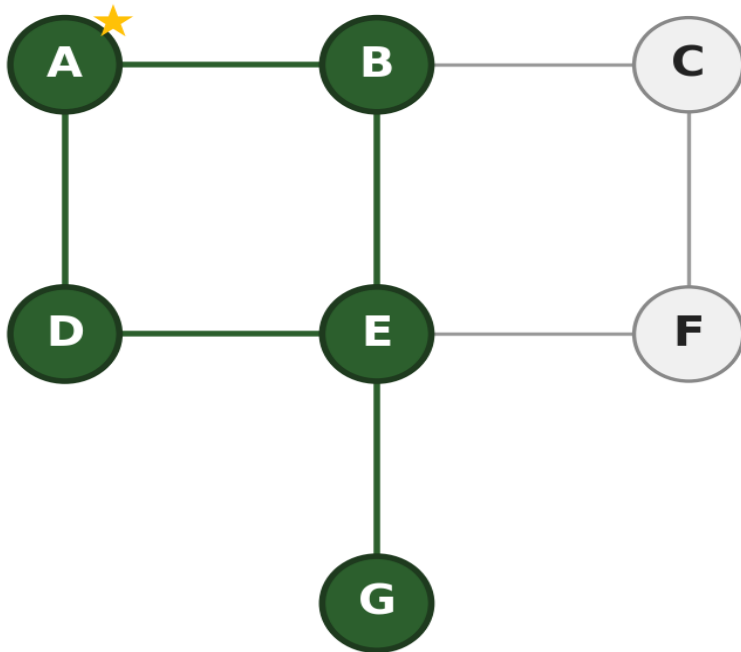
Final: {A, D}, size 2/7

B: $1/3 < 0.4$ X | E: $1/4 < 0.5$ X | No new activations → CASCADE ENDS

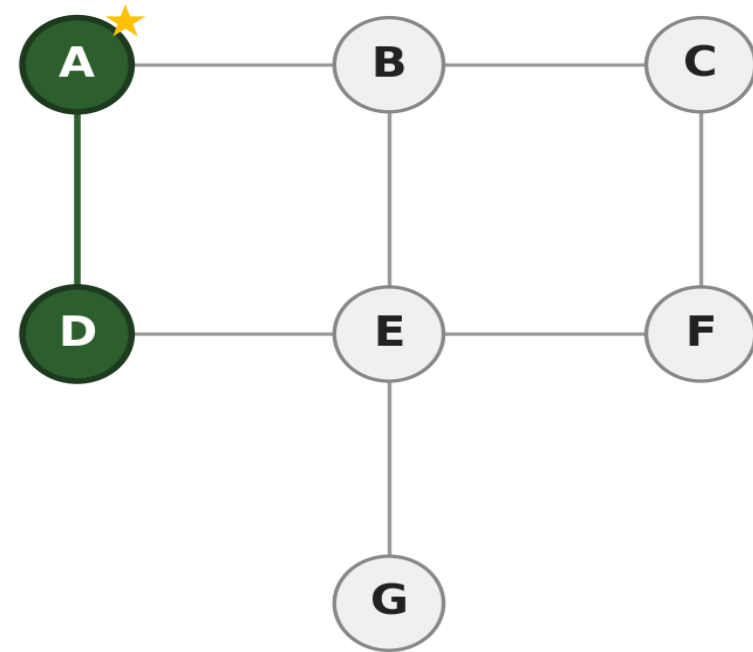
Same Graph, Same Seed, Very Different Outcomes

IC reached 5/7; LT reached only 2/7. Why the gap?

IC Outcome (one realization)



LT Outcome



Cascade size = 5/7 — stochastic (depends on coin flips)

Cascade size = 2/7 — deterministic (depends on thresholds)

The key insight: IC needs *one good roll* to keep going. LT needs *enough committed neighbors* to convert. Different physics → different cascades.

PART 5

Epidemic Models — SIR & SIS

Why a Different Family of Models?

IC and LT are great for one-shot adoption. But many phenomena are not one-shot.

IC / LT

"Once active, always active"

- Adoption is permanent
- Two states: active / inactive
- Cascade ends at some point
- Time index is discrete rounds
- Examples: bought a product, signed up for a service

EPIDEMIC

"State can change over time"

- Nodes can recover, become re-susceptible, etc.
- Multiple compartments: S, I, R (and more)
- Infection has duration — γ governs recovery rate
- Continuous time (ODE) or discrete (network sim)
- Examples: flu, COVID, the spread of a hashtag

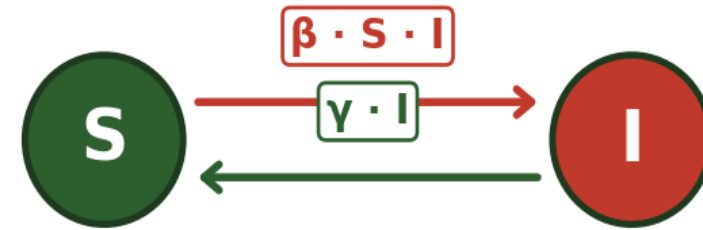
The Two Classic Compartmental Models

SIR and SIS — same basic idea, different recovery dynamics

SIR — Recovered are immune



SIS — Recovered become susceptible again



Transmission rate

How fast S contacts spread infection.
Higher $\beta \rightarrow$ faster outbreak.



Recovery rate

How fast I individuals recover.
 $1/\gamma$ is the average infectious period.



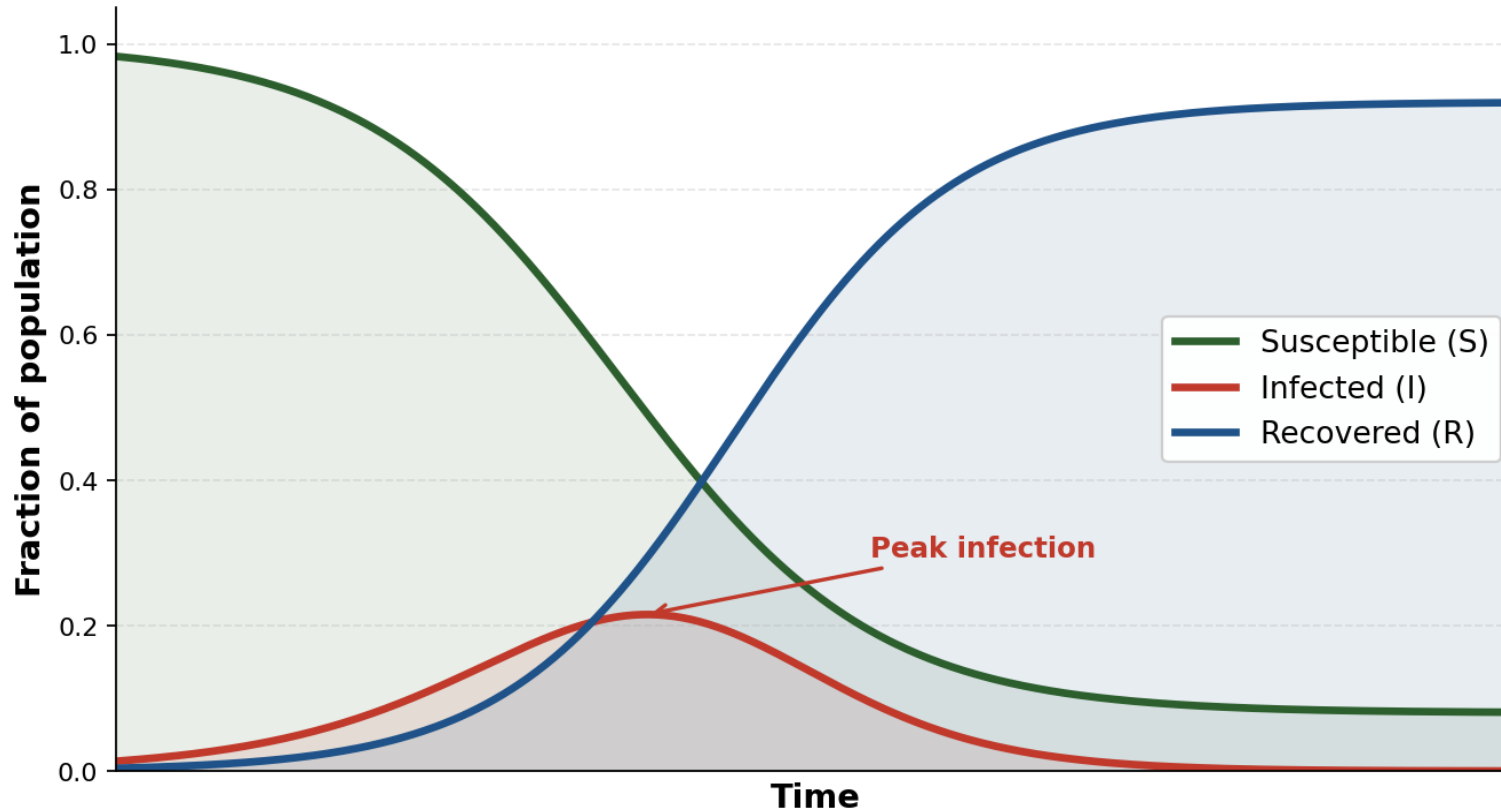
Basic reproduction #

$R_0 = \beta / \gamma$. If $R_0 > 1$, outbreak grows;
if $R_0 < 1$, it dies out.

SIR: The Classic Outbreak-and-Burnout Curve

Susceptibles deplete, infections peak, then everyone has either recovered with immunity or escaped infection

SIR Model — Classic Epidemic Curves



READ THE CURVES

S declines monotonically
as people get infected.

I rises and falls
infections happen, then recovery wins.

R rises monotonically
permanent immunity once recovered.

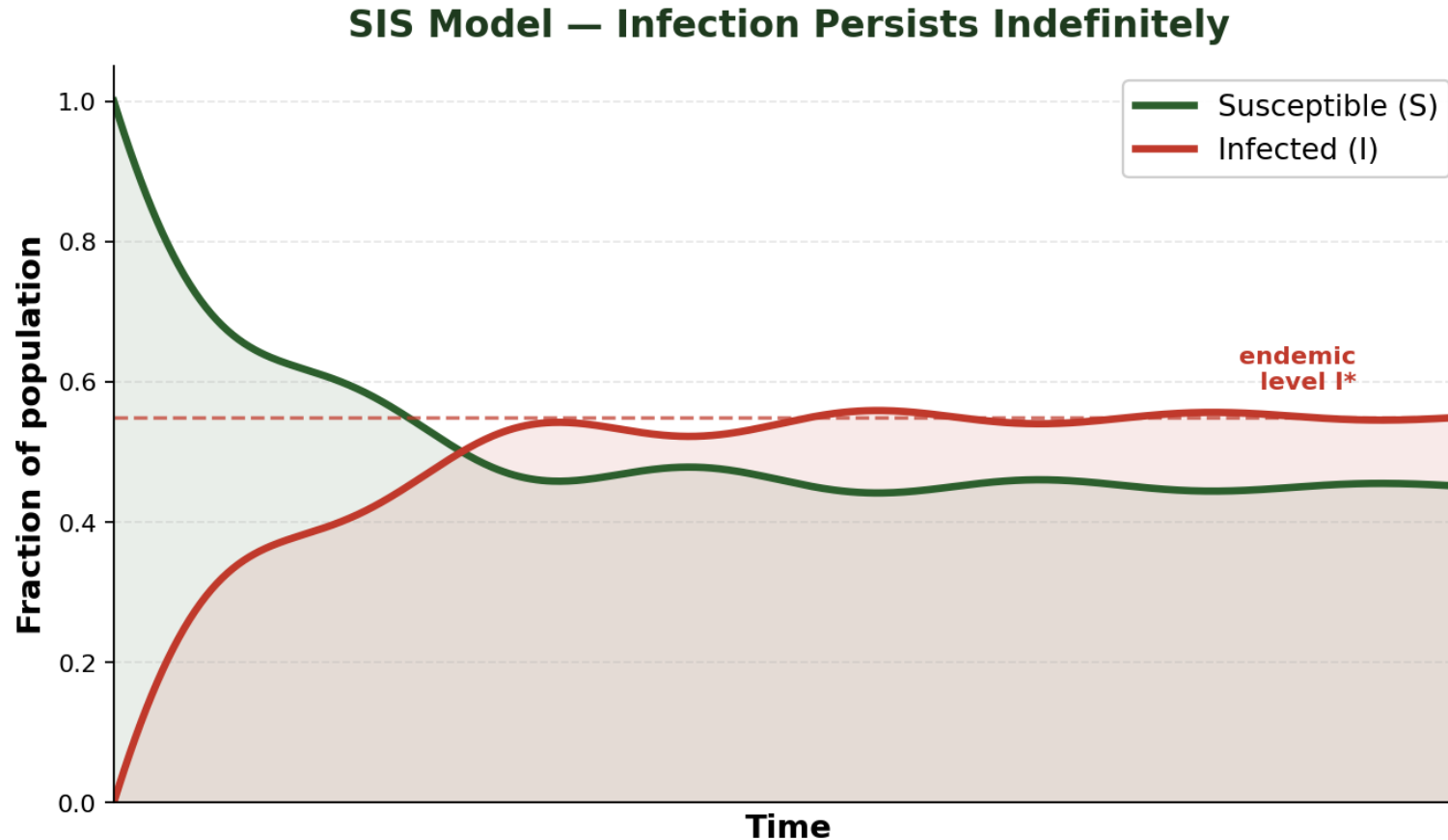
Final: $S^\infty + R^\infty = 1$, $I^\infty = 0$.

Use SIR for:
measles, COVID-19 (with caveats), one-shot
misinformation events

Bottom-line $R_0 > 1 \rightarrow$ outbreak grows; the ultimate cascade size depends on R_0 and network topology.

SIS: When Recovery Doesn't Mean Immunity

Infection persists indefinitely at an endemic equilibrium



READ THE CURVES

Infection settles at I^*
instead of dying out.

S and I oscillate briefly
then reach steady-state balance.

$I^* > 0$ if $R_0 > 1$

Use SIS for:

common cold, computer viruses, recurring fashion trends, evergreen rumors

Bottom-line the persistent endemic level is unique to SIS — SIR cannot produce this shape.

Connecting Epidemics Back to Network Structure

Why structural network analysis matters for epidemic prediction

■ Degree distribution determines super-spreaders

- In scale-free networks, a few high-degree hubs disproportionately drive transmission. Targeted vaccination of these nodes can break an outbreak more efficiently than random vaccination.

■ Average distance determines how fast the wave travels

- Small-world networks (low diameter) let outbreaks reach the entire population in a handful of hops. This is why airport networks and online social platforms accelerate epidemics.

■ Community structure traps infection in clusters

- Strong communities slow spread between groups; bridge nodes (high betweenness) are the gatekeepers. Removing or vaccinating bridges is high-leverage.

■ Centrality measures (week 4) double as risk scores

- Degree centrality flags nodes likely to be infected first; betweenness centrality flags nodes likely to spread to many others.

PART 6

Influence Maximization

The Influence Maximization Problem

Kempe, Kleinberg & Tardos, 2003 — the founding paper

THE PROBLEM

Given a graph G , a diffusion model (IC or LT), and a budget k , find a seed set $S_0 \subset V$ with $|S_0| = k$ that maximizes the expected cascade size $\sigma(S_0)$.

INPUT

Graph G + diffusion model
(IC or LT, plus parameters)

BUDGET

Integer k
(typically much smaller than $|V|$)

GOAL

Pick k nodes that maximize
 $E[\text{cascade size}]$

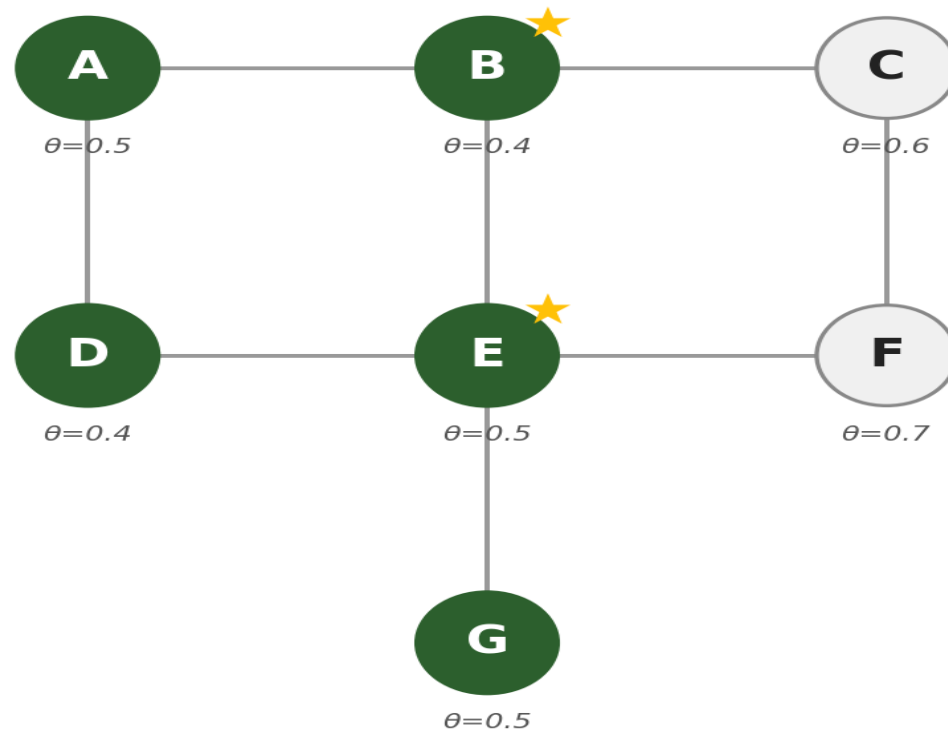
THE CATCH

Choosing the optimal seed set is NP-hard. Brute-force over all $\binom{|V|}{k}$ subsets is infeasible. We need a smart approximation.

Why Picking Top-Degree Nodes Fails

The naïve heuristic is intuitive but wrong

Naïve: Pick top-2 highest-degree nodes



Seeds {B, E} (degrees 3, 4) Cascade = 5/7 ✗ Inactive: {C, F}

PROBLEM: DEADLOCK

Pick the top-2 highest-degree nodes:

→ **Seeds = { E, B }**

→ **LT cascade size = 5/7**

WHY IT'S STUCK:

C needs $\theta = 0.6$, but only B is in its neighborhood (B, F). Pressure stuck at 0.5.

F needs $\theta = 0.7$, but only E is in its neighborhood (C, E). Pressure stuck at 0.5.

C waits for F. F waits for C.

Neither E nor B can break the deadlock — adding more high-degree nodes won't help.

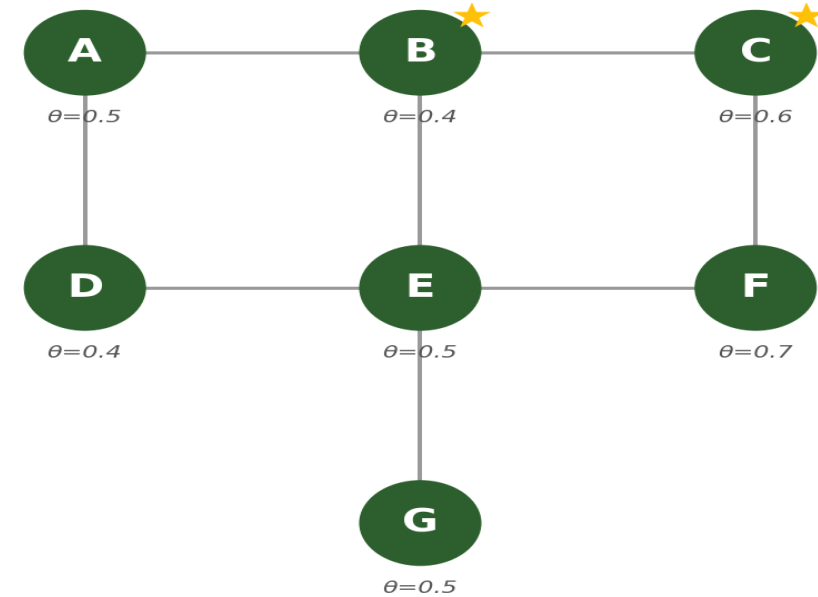
The Greedy Algorithm — and Why It Wins Here

Pick one seed at a time, choosing whichever ADDS the most new coverage

GREEDY ALGORITHM

```
S ← ∅  
for i = 1 to k:  
    pick v* ∈ V \ S that maximizes  
        marginal gain  $\sigma(S \cup \{v\}) - \sigma(S)$   
    S ← S ∪ {v*}  
return S  
  
//  $\sigma(S)$  = expected cascade size with seeds S  
// marginal gain = how many EXTRA  
// nodes v activates that S didn't already
```

Greedy: Pick best single seed, then add best second



Seeds {B, C} Cascade = 7/7 (everyone!) ✓ Greedy beats degree heuristic

Naïve top-degree {E, B} → 5/7 | Greedy {B, C} → 7/7 (B first; then C — the seed that breaks the C/F deadlock)

Why Greedy Works: The Submodularity Property

Diminishing returns is the deep reason greedy is provably good

SUBMODULARITY = DIMINISHING RETURNS

In plain English: the more you've already added, the less each new addition helps.

Formally: if $S \subset T$, then $\sigma(S \cup \{v\}) - \sigma(S) \geq \sigma(T \cup \{v\}) - \sigma(T)$.



Reforestation

Planting your 1st tree increases shade a lot.
Planting your 1,000th tree adds barely any new shade.



TV ads

First ad insertion reaches a new audience. Tenth insertion mostly hits people who already saw it.



Influencer marketing

First influencer reaches their followers. A second influencer with the same followers adds little.

Theorem (Nemhauser et al., 1978): For monotone submodular functions, greedy gives a $(1 - 1/e) \approx 63\%$ approximation to the optimum.

PART 7

Wrap-Up & Project Tie-In

Summary & What Comes Next

From mechanics to optimization — and forward to your final project

MECHANICS

IC, LT, SIR/SIS

Three mechanics for spreading. IC: stochastic, sender-driven. LT: deterministic, receiver-driven. SIR/SIS: epidemic with recovery.

OPTIMIZATION

Influence Maximization

Picking k seeds to maximize cascade is NP-hard. Greedy algorithm, exploiting submodularity, gives $(1 - 1/e)$ approximation.

INSIGHT

Structure matters

Same graph, same seed → wildly different cascade size depending on the model. Choose the model that matches the phenomenon.

PROJECT TIE-IN

Influence Analysis on your project dataset.

Apply today's ideas: identify candidate seeds with PageRank/PPR (weeks 5–6), simulate IC or LT cascades from them, and compare cascade size against the centrality rankings from earlier weeks. Discuss where degree-based heuristics agree or disagree with cascade-based influence.

NEXT WEEK (Week 10):

Network Robustness & Attack Tolerance — random failures vs targeted attacks; the dual of today's diffusion.